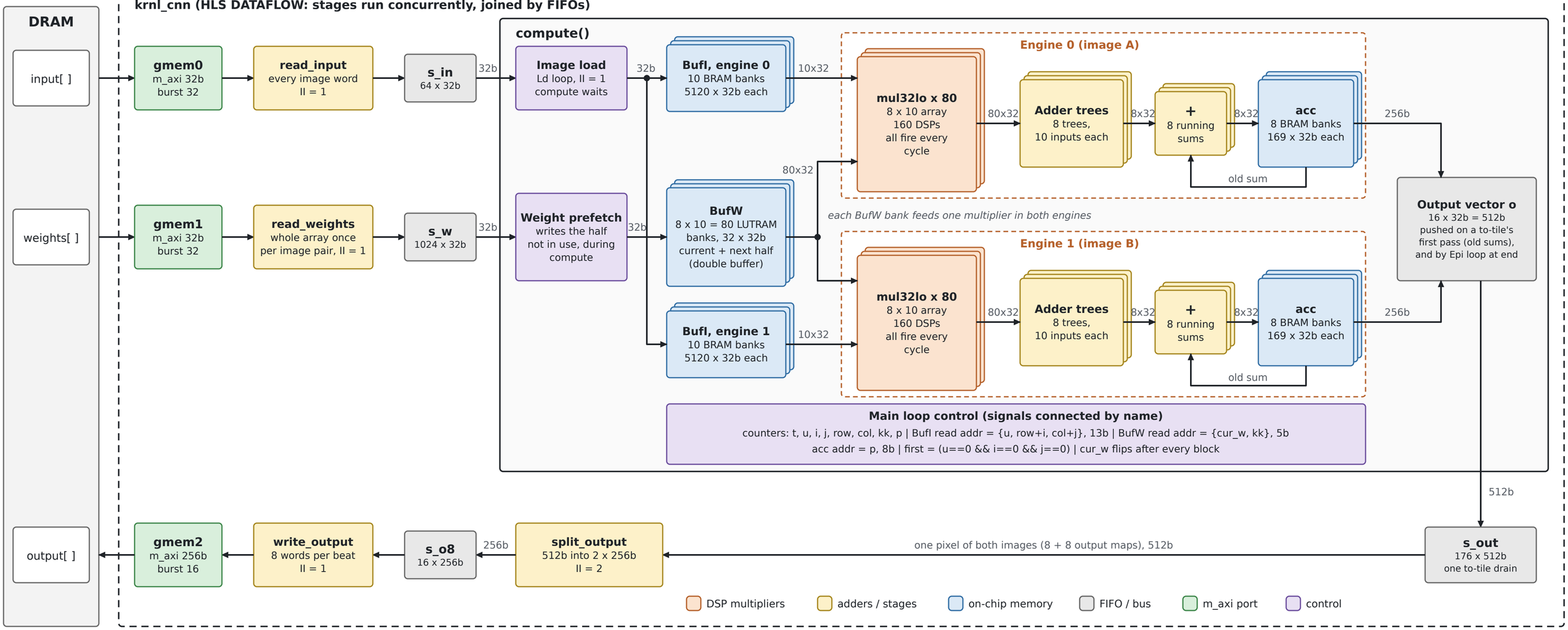
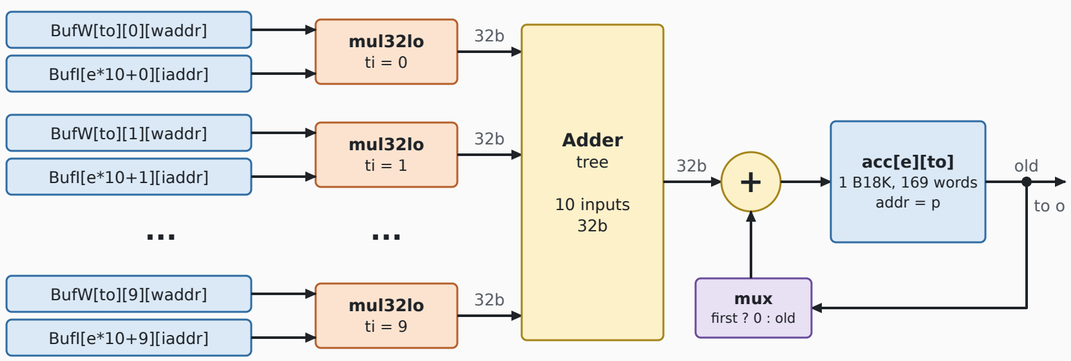
krnl_cnn synthesized kernel datapath

Tm = 8, Tn = 10, Tr = Tc = 13, 2 engines | 160 multipliers per cycle, 2 DSPs each | routed at 200 MHz

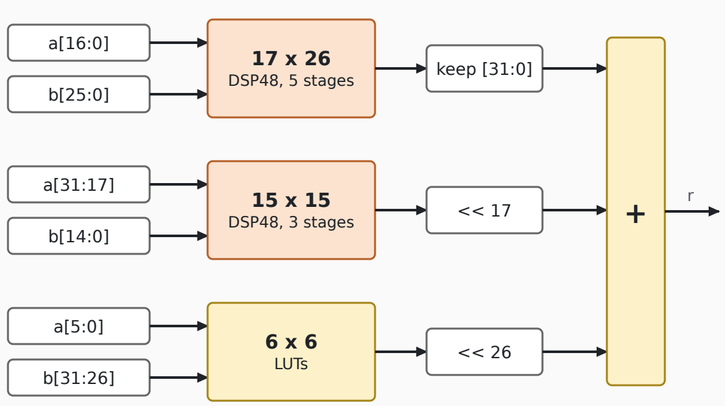


A. One output lane (engine e, output map to): repeated 2 x 8 = 16 times



Main loop: II = 1, iteration latency 10 cycles (pipeline registers not drawn).
One acc word is revisited only 169 iterations later, so its read-add-write spans pipeline stages with no II penalty. In the same cycle a to-tile starts, the old sum leaves through o and is replaced.
Every BufW bank drives this multiplier in both engines; BufI banks are private to one engine.

B. mul32lo: low 32 bits of a x b with 2 DSPs



a = weight, b = input word. Only the low 32 bits are kept, so the a[31:17] x b[31:26] term and the high cross-term bits drop out. A full 32 x 32 multiply costs 3 DSPs; this one costs 2, so 360 DSPs fit 160 multipliers instead of 120.

C. Resources and schedule

Block	Size	Cost
mul32lo	160 instances	320 DSP
read/split/write	4 stages	14 DSP
BufI	20 x 5120 x 32b	280 B18K
acc	16 x 169 x 32b	16 B18K
BufW	80 x 32 x 32b	LUTRAM
Data FIFOs	s_in, s_w, s_out, s_o8	48 B18K
m_axi ports	3	38 B18K
HLS estimate		**334 DSP, 382 B18K**
Routed		**334 DSP, 302 B18K**
Routed timing	200 MHz	WNS +0.486 ns

Per image pair	**cycles**
Ld (load 2 images)	86,403
Main (compute)	486,728
Epi (drain, last pair only)	172